

Analysis of the correlation between knee extension torque and patellar tendon elastic property

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Summary

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Purpose Quadriceps strength and patellar tendon (PT) are directly linked and intimately related to daily activities and lower limb function. However, the correlation between knee extension torque (KT) and PT Young's modulus (E) measured directly is still unknown.

Method We used supersonic shearwave imaging (SSI) to evaluate the elastic property of PT in healthy young men and analysed its correlation with KT. Twenty-two healthy young males were included and both knees were examined. The E of the PT in the dominant and non-dominant legs was assessed by SSI elastography. KT in maximal voluntary isometric contraction was measured with an isokinetic dynamometer.

Result No correlations between KT and PT E were observed in dominant or non-dominant side ($P = 0.458$ and 0.126 , respectively). No significant differences in KT or PT E were observed between both legs ($P = 0.096$ and 0.722 , respectively). Intra-day ICC was rated good (D1 – 0.886 , $P < 0.001$ and 0.88 , $P < 0.001$) and excellent (D2 – 0.928 , $P < 0.001$ and 0.900 , $P < 0.001$) for both legs. Inter-day ICC was rated moderate for both legs (0.651 , $P = 0.016$ and 0.630 , $P = 0.018$, respectively).

Conclusion No significant correlations were found between KT and PT E, suggesting that quadriceps strength is not an accurate predictor for PT mechanical properties in subjects with no specific training engagement. Habitual loading pattern can play a determinant role in PT mechanical properties, regardless of quadriceps strength. Further investigation on SSI acquisition protocols should be conducted to guarantee higher inter-day ICC values.

Introduction

Tendons are important connective tissue structures, responsible for transmitting muscle forces to produce normal joint motion and mechanical torque (Józsa & Kannus, 1997). Its mechanical properties influence movement patterns and are fundamental to guarantee efficient force transmission but also reduced bone–tendon interface stress. The patellar tendon (PT) deserves particular interest for its function, being responsible for transmitting the quadriceps muscle strength, which is fundamental for lower limb movement, locomotion and normal daily activities (Józsa & Kannus, 1997). Patellar tendon properties have been related to the individual loading environment (Couppe et al., 2008; Westh et al., 2008; Kubo et al.,

2010; Seynnes et al., 2013; Zhang et al., 2015), gender and age (Onambele et al., 2007; Westh et al., 2008; Hsiao et al., 2015) and previous existence of any tendon disorder (Zhang et al., 2014; Ooi et al., 2015).

The study of tendons mechanical properties has been of particular interest in the last decades, first performed as an indirect assessment through calculation using B-mode ultrasonography or magnetic resonance imaging (Onambele et al., 2007; Westh et al., 2008; Seynnes et al., 2009, 2013; Svensson et al., 2012) together with dynamometer data. Recently, ultrasound elastography was applied to access PT E in vivo (Kot et al., 2012). In that context, supersonic shear imaging (SSI) rises as a valuable tool for PT evaluation (Kot et al., 2012; Zhang & Fu, 2013; Zhang et al., 2014, 2015; Hsiao et al.,

2015). SSI is able to determine directly the elasticity in a determined region of interest (ROI) without the need of further estimation equations (Kot et al., 2012).

The PT is highly adaptable and resistance training seems to increase tendon stiffness at the same time it promotes quadriceps hypertrophy and strength gains (Reeves et al., 2003; Kongsgaard et al., 2007; Seynnes et al., 2009; Malliaras et al., 2013; Grosset et al., 2014). However, it is controversial if muscle strength can predict tendon stiffness. Training environments that promote strength gains but involve stretch-shortening cycle can reduce PT E (Kubo et al., 2010; Zhang et al., 2015). It is also worth noting that muscle adaptations happen faster than tendons', and therefore, chronic studies trying to correlate increases in muscle strength and PT mechanical property change after training can be limited by the nature of different structures remodelling rate (Wiesinger et al., 2015). In that context, habitual loading environment studies can be considered a valuable tool and were already performed (Couppe et al., 2008; Westh et al., 2008; Kubo et al., 2010; Seynnes et al., 2013; Zhang et al., 2015). Therefore, we hypothesize that in a cross-sectional study with subjects randomly selected, PT E will not correlate with KT.

The main objective of our study was to analyse the correlation between KT and the PT E measured with SSI, in randomly selected healthy young men with no previous history of knee injury. These findings should provide evidence if PT mechanical properties can be related to quadriceps strength, regardless of previous loading environment. Our secondary aim was to test SSI reliability, analysing the ICC between measures made in the same day and between two separate days. This design would provide further data on SSI reliability on PT evaluation, once it is scarce in the literature (Peltz et al., 2013; Zhang & Fu, 2013; Hsiao et al., 2015).

Materials and methods

Ethics statement

The University Hospital Ethics Committee approved this study (registration numbers 570.945 and 519.230). The experimental procedures were conducted in accordance with the 1964 Declaration of Helsinki. All participants received instructions about the study procedures and provided informed written consent before testing.

Experimental procedure

The study was conducted at the Biomedical Engineering Department at the University. The body weight and height of all subjects were measured. Age and dominant leg were informed. All subjects were submitted to PT ultrasound evaluation and KT measure, repeated in two different days with a minimum 1-week rest between them and both knees were examined in each session.

Subjects

In this cross-sectional study, 23 volunteer members of the University Hospital medical staff were selected and both knees were examined. Inclusion criteria were male sex and age between 25 and 30 years old to eliminate any variation of PT properties due to age or gender. Any clinical history or report of knee pain/injuries, systemic disease or previous knee surgery was considered as exclusion criteria. One of the subjects with reported persistent knee pain during examination was excluded, remaining 22 participants. Their demographic characteristics are presented in Table 1. All subjects were right handed and right leg dominant.

Measurement of patellar tendon elastic modulus

An Aixplorer ultrasound (v.9, Supersonic Imaging, Aix-en-Provence, France) in conjunction with a 60-mm linear-array transducer at 4–15 MHz frequency was used in this study. The ultrasound transducer was positioned at the inferior pole of patella and aligned with the patellar tendon, with light pressure on top of a generous amount of coupling gel. B-mode was used to locate and align the patellar tendon longitudinally. When a clear image of the patellar tendon was captured, the shear wave elastography mode was then activated. The transducer was kept stationary for 5-s during the acquisition of the SSI map. A total of two images were captured for the tendon on each knee for offline analysis.

Scanning of PT was performed with the subject in supine lying and the knee at 30° of flexion (Zhang & Fu, 2013). The knee was supported on a custom-made knee and ankle stabilizer to keep the leg in neutral alignment on the coronal and transverse planes. Prior to testing, the subject was allowed to have a 10-min rest in both positions, to ensure the mechanical properties of patellar tendon were evaluated at resting status. The room temperature was controlled at 20°C. The same experienced operator performed all examinations. The rectangular-shaped region of interesting (ROI) was used to display the PT elasticity information (Fig. 1). An Aixplorer tool namely 'Q-box tracing™' was used to obtain the E values. An area between 5 and 20 mm was traced from the inferior pole of the patella excluding the paratendon (Fig. 1). The mean E value was considered the PT E. Each tendon was measured twice.

Measurement of knee extension torque

The maximal isometric KT was measured with an isokinetic dynamometer (Biodex, Biodex Medical Systems, Shirley, NY,

Table 1 Characteristics of subjects in the study.

Variables	n = 22
Age (years)	28.4 ± 2.2
Height (cm)	179.0 ± 7.4
Weight (kg)	87.9 ± 11.4
BMI (kg m ⁻²)	27.2 ± 2.6

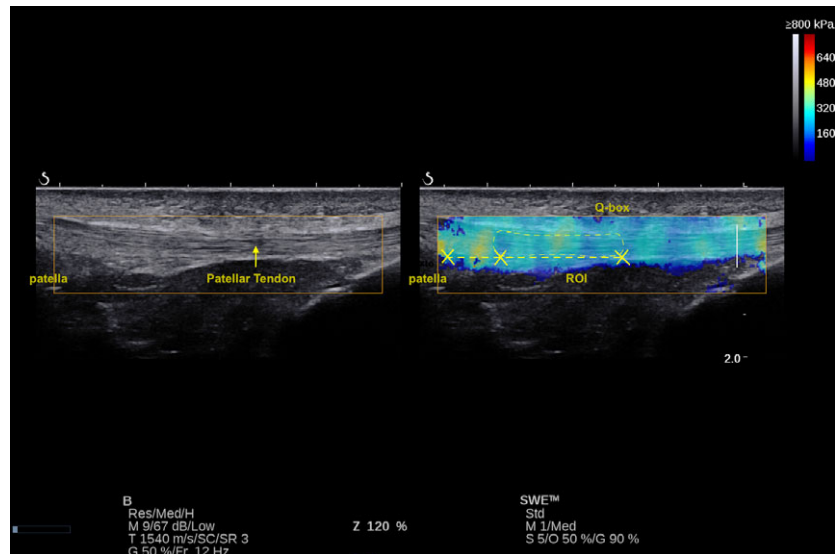


Figure 1 Rectangular ROI with Q-box™ tracing (dashed area) defined between 5 and 20 mm from the patella tip. The colour scale (top right side) represents Young's modulus information and varied from 0 (blue) to 600 kPa (red).

USA) at 70° of knee flexion (Blazevich et al., 2009). Subjects were positioned on the chair with inextensible straps fastened around the waist, trunk and distal part of the thigh. The back-rest inclination and seat translation as well as the dynamometer height were adjusted for each subject, to ensure proper alignment of the rotation axis of the dynamometer with the lateral condyle of the femur. The right knee was fixed to the dynamometer lever arm 5 cm above the lateral malleolus. Settings were recorded for retest reproducibility. After a specific warm-up consisting of two submaximal isometric knee extensions, the subjects performed two 5-s maximal voluntary isometric contractions with 1-min passive rest between trials. Subjects were verbally encouraged to reach maximal effort while a visual feedback of the torque level was provided. The highest peak torque among trials (corrected for gravity) was recorded for analysis.

Statistical analysis

Descriptive data as mean $\pm$ standard deviation (SD) were calculated. After the normality distributions were assumed using the Shapiro–Wilk tests, paired t-tests were used to compare the KT and E of the PT between the dominant and non-dominant sides. Pearson's correlation coefficient was used to investigate the association between the PT E and KT. The dependent measure for analysis was the averaged mean tendon E from two images of the PT taken in each separate day. Intra-operator and SSI reliability were examined using intra-class correlation coefficients (ICC) for measures taken in each day and between separate days. The coefficient of variance (CV) was calculated using the formula $[CV = (\text{standard deviation}/\text{mean}) \times 100\%]$. All reliability coefficients were interpreted as follows: below 0.499 as poor, 0.500–0.699 as moderate, 0.700–0.899 as good, and 0.900–1.000 as excellent (Domholdt, 2000). A P value of 0.05 was considered as significant. All analyses were performed using commercial software GraphPad Prism 5.0 (Graphpad software inc., La Jolla, CA, USA).

Results

Side-to-side differences

No side-to-side differences on the outcome measures were observed in the subjects (Table 2). There were no significant differences on the PT E or KT between the dominant and non-dominant sides ($P > 0.05$).

Correlations

No significant correlations were observed between knee extension torque and Young's modulus in both dominant and non-dominant sides (Figs 2 and 3).

Intra-class coefficient (ICC) values

ICC values for intra-day and inter-day reliabilities on the E of PT were calculated (tables 3 and 4). Intra-day ICC was rated good for both dominant and non-dominant sides in day one (D1 – 0.886, $P < 0.001$ and 0.888, $P < 0.001$, respectively) and excellent in day two (D2 – 0.928, $P < 0.001$ and 0.900, $P < 0.001$, respectively). Inter-day ICC was rated moderate for both dominant and non-dominant sides (0.651, $P = 0.016$ and 0.630, $P = 0.018$, respectively).

Discussion

This study examined the effect of habitual loading on PT mechanical property, based on the correlation between KT

Table 2 Side-to-side differences in patellar tendon Young's modulus and knee extensor torque.

Variables	Dominant side $n = 22$	Non-dominant side $n = 22$	P Value
Thickness (mm)	3.157 ± 0.3	3.232 ± 0.4	0.820
Young's modulus (kPa)	118.8 ± 52.4	137.8 ± 64.9	0.722
Extensor torque (Nm)	305.3 ± 50.3	307.9 ± 54.1	0.096

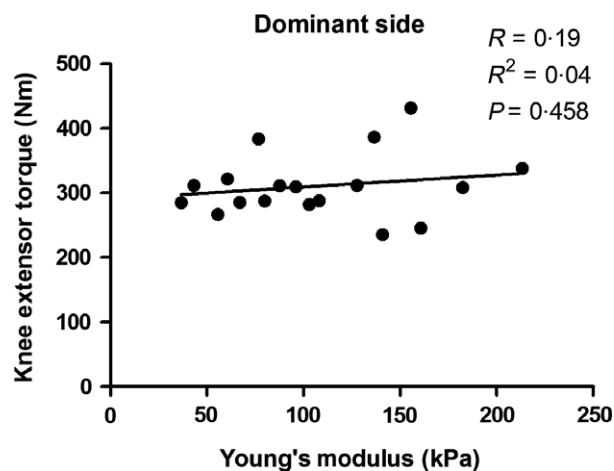


Figure 2 Correlation between knee extension torque and patellar tendon Young's modulus, dominant side

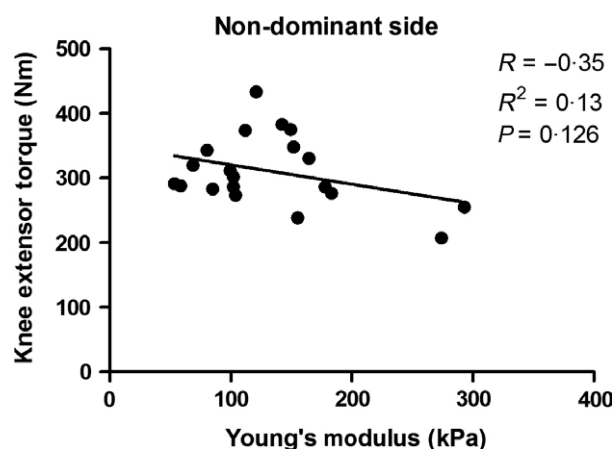


Figure 3 Correlation between knee extension torque and patellar tendon Young's modulus, non-dominant side

and PT E measured by SSI in healthy young males at the dominant and non-dominant legs. Findings indicate that PT E is not correlated to KT, and no significant differences in PT E and KT between legs were detected. We also test the SSI reliability ICC values for measures taken in the same day and in different days. Intra-day ICC was rated good and excellent D1 and D2, respectively, and inter-day ICC was rated moderate.

It is well documented that PT mechanical properties change after long-term resistance training protocols, improving muscle strength and promoting significant increases in PT stiffness and E (Reeves et al., 2003; Kongsgaard et al., 2007; Seynnes et al., 2009; Malliaras et al., 2013; Grosset et al., 2014). Overload exercises demand the tendon unit to work as a rigid force transmitter, therefore the stiffer the better for tendon function in this practice (Wiesinger et al., 2015). However, activities that include high impact and repetitive stretch shortening cycle (i.e. volleyball, basketball) seem to decrease the PT E (Zhang et al., 2015) even though promoting strength and power gains. Also, Kubo et al. (2010) detected higher E

Table 3 Intra-class coefficient (ICC) values for intra- and inter-day reliabilities on the Young's modulus of patellar tendon, dominant side.

	D1 First	D1 Second	D2 First	D2 Second	D1 Mean	D2 Mean
$\bar{x}$	102.01	107.37	123.03	116.87	104.14	119.43
SD	40.99	51.07	77.25	78.40	42.97	74.63
CV%	40.19	47.56	62.79	67.08	41.26	62.49
ICC	0.886		0.928		0.651	
P value	<0.001		<0.001		0.016	
95% CI	0.552–0.921		0.800–0.974		0.094–0.866	

Table 4 Intra-class coefficient (ICC) values for intra and inter-day reliabilities on the Young's modulus of patellar tendon, non-dominant side.

	D1 First	D1 Second	D2 First	D2 Second	D1 Mean	D2 Mean
$\bar{x}$	127.48	122.64	134.41	132.44	123.74	138.31
SD	80.28	62.83	65.01	81.63	59.63	76.14
CV%	62.98	51.24	48.37	61.63	48.19	55.05
ICC	0.888		0.900		0.630	
P value	0.001		<0.001		0.018	
95% CI	0.609–0.968		0.652–0.971		0.065–0.854	

in runners but lower strength when compared to the control group. Therefore, the relation between quadriceps strength and PT properties should not be analysed sole from the resistance training perspective as changes in elastic properties seem to be influenced by sports or training characteristics. Based on these previous evidences, we hypothesize that in subjects randomly selected, PT E would not correlate with KT. Our results showed no correlation between PT E and KT what was coherent with our initial hypothesis. The lack of significant difference in PT E and KT between dominant and non-dominant legs is also consistent with previous findings, as no differential loading environment was detected in volunteers' anamnesis (Couppe et al., 2008).

Reliability of SSI investigation in PT was reported as good to excellent in two previous studies, with ICC values for intra-operator reliability ranging from 0.835 to 0.991 (Zhang & Fu, 2013; Hsiao et al., 2015) and from poor to excellent in one (Peltz et al., 2013). It is worth noting that in the work conducted by Peltz et al. (2013), the examiner was not a trained experimenter. Although the manufacturer states that the technique is 'user-skill-independent', it may have impacted reproducibility. In our study, the examiner had 2 years of previous experience with SSI and our intra-operator ICC values ranged from 0.886 to 0.928 for measures taken in the same day. These results are consistent with the ones previously presented in the literature. A relevant aspect analysed in our work is the examination reproducibility in different time testing points. We determine it by comparing measures from two separate days. The ICC values were rated moderate (0.630 and 0.651) for non-dominant and dominant sides,

respectively. To our knowledge, only Hsiao et al. (2015) have previously addressed this question. Examining the proximal PT in a reduced sample composed of five subjects in two different days with a 1-week interval, they found an intra-rater ICC value ranging from 0.742 to 0.874. Our reproducibility values were relatively lower and comparing both protocols, one important factor could be identified. Their group included a 2-h rest recommendation before the examination. Although our exams were performed in the morning period and subjects reported no previous physical activity in that day, no additional control was made for daily activities as walking long distances or going up or down stairs. This could acutely change PT properties in different days, what would impair reproducibility. Further investigation controlling these parameters and analysing ICC with well-defined pre-examination protocols should be tested.

We recognize some study limitations. First, we used a convenience sample composed of healthy young male volunteers from the staff of our institution, so these findings cannot be generalized to females, other age range or a specific athletic population. Potential confounding factors such as smoking and nutrition were not accounted for, although all the participants were free of systemic or local disease. The leg preference in the present study was self-reported, and no experimental assessment of hand or leg preference was performed. At last, this was a cross-sectional observational study; no causal effects could be analysed. Additional longitudinal trials are warranted for further clarification of the effect of

quadriceps function on PT properties. Furthermore, long-term interventions protocols should provide an insight into how tendon adapts when examined in the light of SSI.

Conclusion

The present study revealed no significant differences in PT E and KT between legs, suggesting that in similar loading environments, side dominance does not affect these parameters. No significant correlations were found between KT and PT E, suggesting that quadriceps strength is not predictor for PT mechanical properties in subjects with no specific training engagement. ICC values from SSI were higher for measures taken in the same day when compared to measures taken in different days. Further investigation on SSI acquisition protocols should be conducted to guarantee higher inter-day ICC values and greater SSI reliability.

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Conflict of interests

The authors declare they have no conflict of interests.

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